

Learning-enhanced Video Compression with Capability beyond VVC

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Abstract—ITU-T VCEG and ISO/IEC MPEG's Joint Video Experts Team (JVET) has issued a Call for Evidence (CfE) to study the advanced video coding technologies that could provide substantially higher compression capability than VVC standard. This paper presents one solution based on Enhanced Compression Model (ECM) by integrating two optimized learning-based coding tools: a Cost Volume-based Deep Reference Frame (CV-DRF) and a Test Time Augmentation-enhanced Neural Network Loop Filter (TTA-NNLF). The CV-DRF employs a cost volume-based video frame interpolation network to synthesize a temporally consistent reference frame that strengthens inter prediction, while the TTA-NNLF adopts a Transformer-based filter to suppress compression artifacts and improve reconstruction quality. When evaluated under the Random Access (RA) configuration on the CfE SDR RA UHD, SDR RA HD, UGC RA test sequences, relative to the CfE anchor VTM-23.11, this method achieves average BD-rate savings of 32.02%, 36.15%, and 38.39% for Y, U, and V components, respectively. These results provide concrete evidence that hybridizing ECM with modern learned modules is a promising direction for next-generation standardization beyond VVC.

Index Terms—Call for Evidence, Versatile Video Coding, Enhanced compression model, Deep reference frame, Neural network loop filter

I. INTRODUCTION

The Joint Video Experts Team (JVET) of ITU-T SG16 WP3 and ISO/IEC JTC 1/SC 29 has developed and finalized the latest video coding standard, namely Versatile Video Coding (VVC)/H.266 [1]. VVC has made significant progresses in terms of coding efficiency, providing approximately a 50% bitrate saving for equivalent perceptual quality relative to the prior standard High Efficiency Video Coding (HEVC)/H.265 [2]. While VVC offers a new level of capability for video compression, the ongoing demand for higher compression efficiency across diverse applications such as UHD/HDR streaming, cloud gaming, and user-generated content motivates investigation beyond the VVC standard. Following the finalization of VVC, JVET continues to explore coding technologies beyond VVC through two complementary research tracks: Neural Network-based Video Coding (NNVC) [3], which explores learning-based compression approaches, and the Enhanced Compression Model (ECM) [4], which develops traditional coding tools.

The ECM track builds upon the conventional block-based framework of VVC by refining core coding tools rather than overhauling the codec structure. For example, ECM incorporates advanced prediction tools, such as decoder-side intra mode derivation (DIMD) [5], which infers the intra prediction mode by computing a histogram of gradients; intra template matching prediction (IntraTMP) [6], which copies the best prediction block from the reconstructed part of the current frame, whose L-shaped template matches the current template; and multi-pass decoder-side motion vector refinement (DMVR) [7], which iteratively refines motion vectors to improve motion compensation. Together with additional techniques summarized in [8], these components enable superior rate distortion performance while preserving compatibility with traditional coding structures. Meanwhile, NNVC explores learning-based modules incorporated into the hybrid coding framework. The NN-based Intra Prediction (NN-Intra) and NN-based In-loop Filter (NNLF) [9] employ specialized deep learning architectures to improve prediction accuracy and artifact reduction, while maintaining compatibility with the existing tools of conventional codecs. Moreover, the Deep Reference Frame (DRF) technique [10] enhances inter prediction by generating synthesized reference frames via multi-scale optical flow estimation and hierarchical feature fusion. Together with other neural tools described in [11], NNVC demonstrates the potential of deep learning to further improve compression efficiency beyond the capability of traditional hybrid codecs.

To study video compression technologies that have more powerful compression performance or additional functionality beyond that of VVC, JVET issued a Joint Call for Evidence (CfE) [12] on video compression beyond VVC at the 39th meeting in Daejeon, in preparation for launching a formal next-generation standardization project. VVC anchors are generated using the VTM-23.11 [13] software. JVET received four technical responses [14]–[17] covering compression improvements under runtime-unconstrained and runtime-constrained settings, most based on VTM or ECM and incorporating learning-based coding techniques. Subjective viewing tests indicate clear visual improvements over VVC [18].

In this paper, we present our CfE solution [16] and provide independent analysis and discussions. We integrate two optimized neural network-based tools into ECM-17.0 [19], namely a Cost Volume-based Deep Reference Frame (CV-

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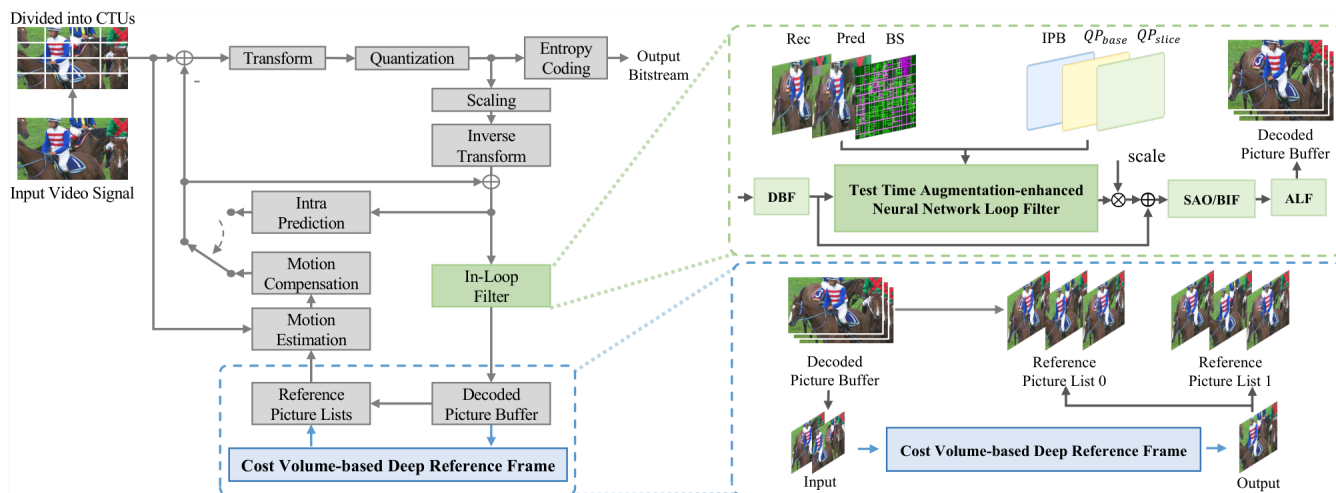


Fig. 1. The overall framework of the proposed CfE response solution.

DRF) and a Test Time Augmentation-enhanced Neural Network Loop Filter (TTA-NNLF). We submitted corresponding test results for the SDR RA UHD, SDR RA HD, and UGC RA test cases in the CfE response. Relative to the CfE test anchor VTM-23.11, this solution delivers average BD-rate reductions of 32.02%, 36.15%, and 38.39% for the three test cases under RA configuration on the Y, U, and V components, respectively.

II. OVERVIEW OF CFE RESPONSES BEYOND VVC

A. JVET-AN0085

JVET-AN0085 [14] is built entirely on ECM-18.0. Three test settings are evaluated: (1) the default ECM configuration; (2) ECM with constrained complexity (partitioning restricted to reduce encoder runtime); and (3) ECM with constrained complexity plus the LOP6 filter. For SDR RA UHD, SDR RA HD, and UGC RA test category, under the RA configuration, the best-performing setting (Test 1) achieves average BD-rate reductions of 25.15%, 30.29%, and 33.72% over VTM-23.11 on the Y, U, and V components, respectively.

B. JVET-AN0198

JVET-AN0198 [16] corresponds to the method presented in this paper. The design and integration details are provided in Section III. For SDR RA UHD, SDR RA HD, and UGC RA test category, this response achieves average BD-rate reductions of 32.02%, 36.15%, and 38.39% over VTM-23.11 under the RA configuration on the Y, U, and V components, respectively.

C. JVET-AN0212

JVET-AN0212 [17] is based solely on NNVC-14.1. The software includes neural-network tools such as NNLF, NN-Intra, DRF, GOP-based RPR, time-distortion optimization (TDO), and filter restriction by hierarchical depth. Under the RA configuration, the best test configuration achieves average BD-rate reductions of 13.18%, 19.12%, and 15.79% on the Y, U, and V components over VTM-23.11 for SDR RA UHD, SDR RA HD, and UGC RA test category.

D. JVET-AN0267

JVET-AN0267 [15] provides two responses implemented in an alternative codebase called NextSoftware2, derived from VTM-20.0 with extensive clean-up. Several ECM tools offering favorable runtime–efficiency trade-off are incorporated into this baseline. The two submissions differ only in whether a VLOP-like NNLF is enabled. The better submission attains average BD-rate reductions of 20.96%, 36.92%, and 38.07% on the Y, U, and V components over VTM-23.11 for SDR RA UHD, SDR RA HD, and UGC RA test category.

E. Visual Improvements over VVC

JVET-AN0361 [18] reports the results of the CfE response viewing tests and provides MOS-rate curve comparisons of different responses for several representative sequences. For most categories, significant compression improvements can be observed when comparing the proposed methods with the VTM-23.11 anchor.

III. METHODOLOGY

This response of JVET-AN0198 [16] is based on ECM 17.0 [19], with the addition of two optimized neural network-based coding tools, CV-DRF and TTA-NNLF. The CV-DRF tool inserts a virtual frame synthesized by a video frame interpolation model into the Reference Picture List (RPL), making it available for motion estimation and compensation to enhance inter prediction. The TTA-NNLF is an in-loop neural filter that suppresses compression artifacts in reconstructed output frames and improves the quality of reference pictures. The overall framework of the CfE response is illustrated in Figure 1.

A. Cost Volume-based Deep Reference Frame

Building on VVC, ECM introduces a broad set of inter coding techniques, with representative techniques including template matching (TM), local illumination change (LIC), non-adjacent merge candidates, overlapped block motion compensation (OBMC), and so on. These mainly refine motion

estimation, motion vector candidate derivation, and motion compensation, yet comparatively limited effort targets the quality of reference frames. To address this limitation, we integrate a cost volume-based deep reference frame generation method into ECM that synthesizes an additional reference frame closely approximating the current frame, thereby strengthening inter prediction, and the framework of this tool is illustrated in Figure 1.

Under the RA configuration with bidirectional prediction, the CV-DRF network takes two reconstructed frames from the decoded picture buffer that are temporally symmetric around the current picture and have the smallest available Picture Order Count (POC) distance. It interpolates an additional reference frame, which is then inserted as the second entry in both RPL 0 and RPL 1 assigned the current picture's POC. The tool is enabled only for higher temporal layers, specifically layers 3 to 5, with input selection adapted to the layer.

Different from the task-oriented flow-based generation network used in [10], we adopt the cost volume-based All-Pairs Multi-Field Transforms (AMT) framework of [20], which builds on Recurrent All-Pairs Field Transforms (RAFT) [21]. In this framework, a correlation encoder first maps the input frames into a pair of dense feature representations to construct bidirectional correlation volumes, also known as cost volumes. Meanwhile, a context encoder generates the initial intermediate feature, bilateral flows, and pyramid features for progressive warping. The cost volumes are computed as the dot-product similarities between all pairs of feature vectors extracted by the correlation encoder. After constructing the bidirectional correlation volumes, we query the correlation feature maps using the estimated bilateral flows through a scaled lookup operation, which addresses the coordinate mismatch between visible and invisible frames. During the update stage at each spatial level, update blocks jointly predict the residuals of the bilateral flow fields and the interpolated intermediate features based on the retrieved bidirectional correlations. Finally, a multi-field refinement operator interpolates the target frame using multiple flow groups at the finest scale.

B. Test Time Augmentation-enhanced Neural Network Loop Filter

ECM advances a well-optimized in-loop filtering pipeline beyond VVC. It strengthens Adaptive Loop Filtering (ALF) and Cross-Component ALF (CCALF) and introduces Bilateral Filters (BIF) for the luma and chroma channels together with Cross-Component Sample Adaptive Offset (CCSAO) [19]. These enhancements substantially improve artifact suppression and increase overall coding efficiency. However, the existing filters are primarily traditional signal-processing designs or models trained offline. To further improve the in-loop stage, a Test Time Augmentation-enhanced Neural Network Loop Filter (TTA-NNLF) is incorporated into ECM.

As illustrated in Figure 1, TTA-NNLF operates in parallel with the Deblocking Filter (DBF). The residual between the TTA-NNLF and DBF outputs is scaled and then added to the DBF result to form an enhanced picture, which is then

passed sequentially through SAO/BIF and finally ALF for further refinement at both encoder and decoder. We adopt the Transformer-based filter of [22], where a transformer encoder block is introduced to the end of each backbone block in HOP filter defined in [23]. The inputs include a reconstructed YUV frame (Rec), where the U and V planes are spatially upsampled to the Y resolution, together with auxiliary signals: prediction samples (Pred), boundary strength (BS), quantization parameters (QP_{base} , QP_{slice}), and prediction types. In addition, we introduce Test-Time Augmentation (TTA) into the codec for network inference, which is commonly used in computer vision tasks to improve inference accuracy [24], [25]. The output of TTA-NNLF is then added to the reconstructed YUV frame (Rec) to obtain the filtered result.

C. SADL Implementation in ECM

To ensure cross-platform interoperability and bit-exact reproducibility, we use the Small Ad-hoc Deep-Learning Library (SADL) framework [26] developed by JVET for integer neural network inference. SADL is a lightweight, header-only library that integrates easily with codec. To support the two tools described above, we introduce new operators including InstanceNormalization, LayerNormalization, Div, Sigmoid, and Softmax into the latest release SADL-14.0 [27]. In addition, we enhance the functionality of existing SADL operators such as GridSample, Resize, Concat, and Slice, and provide SIMD acceleration for int32 inference on a subset of operators.

IV. EXPERIMENTS

A. Training

We train the video frame interpolation network used in CV-DRF on a compressed subset of the Vimeo-90K triplet dataset [28]. The first and third frames of each triplet are encoded under the All Intra (AI) configuration with Quantization Parameter (QP) 22, 27, 32, and 37, while the original middle frame serves as the ground truth frame. Training follows a QP-Progress strategy, starting with low QP data to learn motion and then gradually adding high QP data to enhance artifact suppression, which yields better robustness across reconstruction qualities. The model is trained for 300 epochs on four RTX 3090 GPUs with a batch size of 32, and a cosine-annealing schedule that decays the learning rate from 1.5×10^{-4} to 1.5×10^{-5} . The data augmentation and training loss follow those used in [20].

B. Experimental Results and Analysis

We evaluate the proposed method on the CfE test sequences, including SDR RA UHD, SDR RA HD, and UGC RA, as well as on the VVC Class B, C, and D. For the CfE evaluation, the bitrate points are set strictly according to the CfE test conditions [12]. For the VVC test sequences, the QP settings follow the ECM common test conditions [29]. BD-rate [30] is employed to assess coding efficiency. As shown in Table I, compared to CfE anchor VTM-23.11, under the RA configuration, for the CfE SDR RA UHD, SDR RA HD, and UGC RA sequences, our method achieves

TABLE I
PERFORMANCE OF THE PROPOSED METHOD COMPARED WITH VTM-23.11 UNDER RA CONFIGURATION

Class	Sequence	BD-rate and Complexity over VTM-23.11				
		Y	U	V	EncT(CPU)	DecT(CPU)
CfE SDR RA UHD	CrowdRun	-29.19%	-30.99%	-26.67%	4844.70%	3906276.21%
	DrivingPOV3	-35.98%	-34.59%	-56.16%	10176.63%	4046930.46%
	FireDance	-39.21%	-38.84%	-35.50%	6576.29%	4105405.48%
	HallwayScene	-45.10%	-57.51%	-53.38%	9466.73%	4225048.10%
	Average	-37.37%	-40.48%	-42.93%	7443.25%	4069294.31%
CfE SDR RA HD	DucksTakeOff	-20.31%	-64.46%	-76.01%	3811.58%	3807375.62%
	DrivingPOV4	-32.83%	-29.64%	-36.73%	9174.65%	4010982.38%
	Seeking	-28.08%	-28.95%	-13.78%	5864.17%	4585305.86%
	Umbrella	-30.95%	-27.54%	-41.31%	4675.33%	3989596.93%
	Average	-28.04%	-37.65%	-41.96%	5564.53%	4088306.55%
CfE UGC RA	Camellia	-33.78%	-23.20%	-26.09%	7495.85%	3135669.01%
	Hobby-w5xz-backpack	-32.43%	-23.01%	-35.81%	11167.02%	4541791.88%
	NightLandscapeRunning	-30.98%	-41.94%	-35.53%	7788.43%	3889414.23%
	Sports-76a2-iceball	-40.16%	-26.56%	-34.40%	15393.77%	4547589.41%
	VerticalVideo-3709-snow	-26.13%	-45.22%	-34.39%	11725.63%	4126232.74%
	VerticalVideo-3d96-walk	-23.21%	-33.71%	-31.67%	11712.54%	4123926.60%
	Average	-31.11%	-32.27%	-32.98%	10549.29%	4030397.89%
Average (CfE UHD/HD/UGC)		-32.02%	-36.15%	-38.39%	7953.88%	4057979.41%
Class B	MarketPlace	-27.64%	-39.19%	-32.59%	2735.41%	2461139.52%
	RitualDance	-29.86%	-37.66%	-34.16%	2773.25%	2710775.19%
	Cactus	-33.00%	-40.20%	-34.60%	2971.91%	2728704.46%
	BasketballDrive	-33.72%	-35.79%	-33.98%	2398.42%	2659965.04%
	BQTerrace	-35.28%	-47.65%	-43.74%	3560.83%	3030951.99%
	Average	-31.90%	-40.10%	-35.81%	2863.55%	2712240.80%
Class C	BasketballDrill	-35.59%	-35.15%	-36.69%	2231.94%	2257543.36%
	BQMall	-35.10%	-30.85%	-31.40%	2845.66%	2589585.96%
	PartyScene	-33.34%	-26.53%	-23.77%	2214.69%	1912762.85%
	RaceHorses	-29.81%	-23.45%	-30.92%	1971.27%	1858319.41%
	Average	-33.46%	-28.99%	-30.70%	2294.73%	2135069.33%
Class D	BasketballPass	-32.33%	-31.28%	-30.22%	1897.28%	1335731.35%
	BQSquare	-41.30%	-36.48%	-42.78%	2216.71%	1763016.30%
	BlowingBubbles	-30.69%	-18.37%	-17.72%	2097.68%	1439111.04%
	RaceHorses	-31.89%	-31.16%	-33.98%	1823.89%	1091931.38%
	Average	-34.05%	-29.32%	-31.17%	2002.83%	1386967.46%
Average (Class B-D)		-33.04%	-33.37%	-32.81%	2396.30%	2049910.70%

average BD-rate reductions of 32.02%, 36.15%, and 38.39% on the Y, U, and V components, respectively. For the VVC Class B, C, and D test sequences, the corresponding averages are 33.04%, 33.37%, and 32.81%, respectively. While neural network-based tools deliver substantial coding gains, they also introduce high computational complexity as shown in Table I, which is challenge in practical use.

C. Ablation Studies

To analyze the individual contributions of the two neural-network tools and to examine their interaction under joint enabling, we conduct an ablation study on Classes C, and D under the ECM CTC RA configuration [29]. As reported in Table II, relative to the ECM-17.0 baseline, enabling CV-DRF alone yields average BD-rate gains of 3.77%, 3.34%, and 3.37% on the Y, U, and V components, respectively. Enabling TTA-NNLF alone provides BD-rate gains of 6.64%, 4.61%, and 6.35% on the Y, U, and V components, respectively. When both tools are enabled, the achieved gain is about 1% smaller than the sum of the two individual gains, indicating partial overlap rather than perfect additivity. This overlap arises because both modules optimize the reference pictures: CV-DRF synthesizes a reference frame that is temporally closer to the current frame, while TTA-NNLF improves the fidelity of the reconstructed reference frames. Both effects reduce inter

TABLE II
ABLATION STUDY OF NEURAL NETWORK-BASED TOOLS

Tools	BD-rate over ECM-17.0								
	Class C			Class D			Average		
	Y	U	V	Y	U	V	Y	U	V
(a) CV-DRF only	-3.79%	-3.86%	-3.83%	-3.76%	-2.82%	-2.91%	-3.77%	-3.34%	-3.37%
(b) TTA-NNLF only	-6.23%	-4.12%	-5.62%	-7.06%	-5.10%	-7.08%	-6.64%	-4.61%	-6.35%
(c) Both	-9.13%	-7.45%	-8.63%	-9.40%	-8.11%	-9.45%	-9.27%	-7.78%	-9.04%

prediction error and thereby facilitate more accurate motion estimation and motion compensation.

V. CONCLUSIONS

This paper presents a CfE solution that augments the ECM with two optimized neural network-based coding tools: CV-DRF and TTA-NNLF. Under the RA configuration, relative to VTM-23.11, for the SDR RA UHD, SDR RA HD, UGC RA test sequences, this framework achieves average BD-rate reductions of 32.02%, 36.15%, and 38.39% on the Y, U, and V components, respectively. Moreover, this solution achieves remarkable compression performance for runtime-unconstrained track in the CfE evaluation. Overall, the results substantiate that coupling ECM's hand-crafted tools with modern learned components is an effective route for video coding beyond VVC. At the same time, the introduced computational complexity is a challenge issue in further study.

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